

Programming Paradigms

BSc Computer Science - Durham University

Matthew Emerson

October 2026

Contents

1	Module Introduction	1
1.1	Assessment	1
1.2	Timetable	1
2	Systems Programming	1
2.1	Version Control and UNIX	1
2.2	The Basics of C	1
2.3	Control Flow and Functions	1
2.4	Data Types and Structures	1
2.5	Advanced Bash Scripting	1
2.6	Pointers	1
2.7	Memory Management	1
2.8	Debugging	1
2.9	Scope and Recursion	1
2.10	Libraries	1
2.11	stdlib	1
2.12	C++	1
3	Functional Programming	1
4	Object Oriented Programming	2
4.1	Java Fundamentals	2
4.2	OOP with Classes and Objects	2
4.3	Encapsulation	2
4.4	Inheritance	2
4.5	Abstraction with Abstract Classes	2
4.6	Abstraction with Interfaces	2
4.7	Polymorphism	2
4.8	Java Collections	2
4.9	Best Practises	2
I	Appendix	3
5	Bibliography	3
5.1	Module Professors	3
5.2	References	3
5.3	List of Figures	3

1 Module Introduction

Programming Paradigms has the module code COMP2221 and is led by Dr Stuart James.

1.1 Assessment

This module is assessed through one piece of summative coursework and two final exams, these can be seen in Table 1. **Final dates to be confirmed.**

Title	Opening	Due	Feedback	Professor	Weight
Systems Programming	31/10/2026	11/12/2026	08/01/2027	Dr Stuart James	50%
Object Oriented & Functional	May Exam	n/a	n/a	Ivkovic & Moses	50%

Table 1: Assessments for Programming Paradigms.

1.2 Timetable

This module is taught through two one-hour lectures and one two-hour computer class each week as seen below in Table 2.

Day	Time	Class Type	Location	Lecturer
Monday	11:00 - 12:00	Lecture	CLC202	Dr Stuart James
Monday	16:00 - 18:00	Computer Class	CB-LG001	n/a
Friday	14:00 - 15:00	Lecture	CLC202	Dr Stuart James

Table 2: Michaelmas Term Timetable for Programming Paradigms.

2 Systems Programming

This section is taught by Dr Stuart James. **Check for recommended reading.**

2.1 Version Control and UNIX

2.2 The Basics of C

2.3 Control Flow and Functions

2.4 Data Types and Structures

2.5 Advanced Bash Scripting

2.6 Pointers

2.7 Memory Management

2.8 Debugging

2.9 Scope and Recursion

2.10 Libraries

2.11 stdlib

2.12 C++

3 Functional Programming

This section is taught by Dr Billy Moses Jr.. **Check for recommended reading.**

4 Object Oriented Programming

This section is taught by Dr Mladen Ivkovic. There is no recommended reading for this topic.

- 4.1 Java Fundamentals
- 4.2 OOP with Classes and Objects
- 4.3 Encapsulation
- 4.4 Inheritance
- 4.5 Abstraction with Abstract Classes
- 4.6 Abstraction with Interfaces
- 4.7 Polymorphism
- 4.8 Java Collections
- 4.9 Best Practises

Part I

Appendix

5 Bibliography

5.1 Module Professors

Module Lead: Doctor Stuart James

- Focus: Artificial intelligence, computer vision, human computer interaction, and digital humanities.
- Email: stuart.a.james@durham.ac.uk
- Office Hours: Email to make an appointment.

Lecturer: Doctor Billy Moses Jr.

- Focus: Algorithms and biologically inspired computing.
- Email: william.k.moses-jr@durham.ac.uk
- Office Hours: Email to make an appointment.

Lecturer: Doctor Mladen Ivkovic

- Focus: Performance prediction and optimisation.
- Email: mladen.ivkovic@durham.ac.uk
- Office Hours: Email to make an appointment.

5.2 References

5.3 List of Figures